

Posterior Sampling for Solar Snapshot Spectroscopy with Score-Based Priors

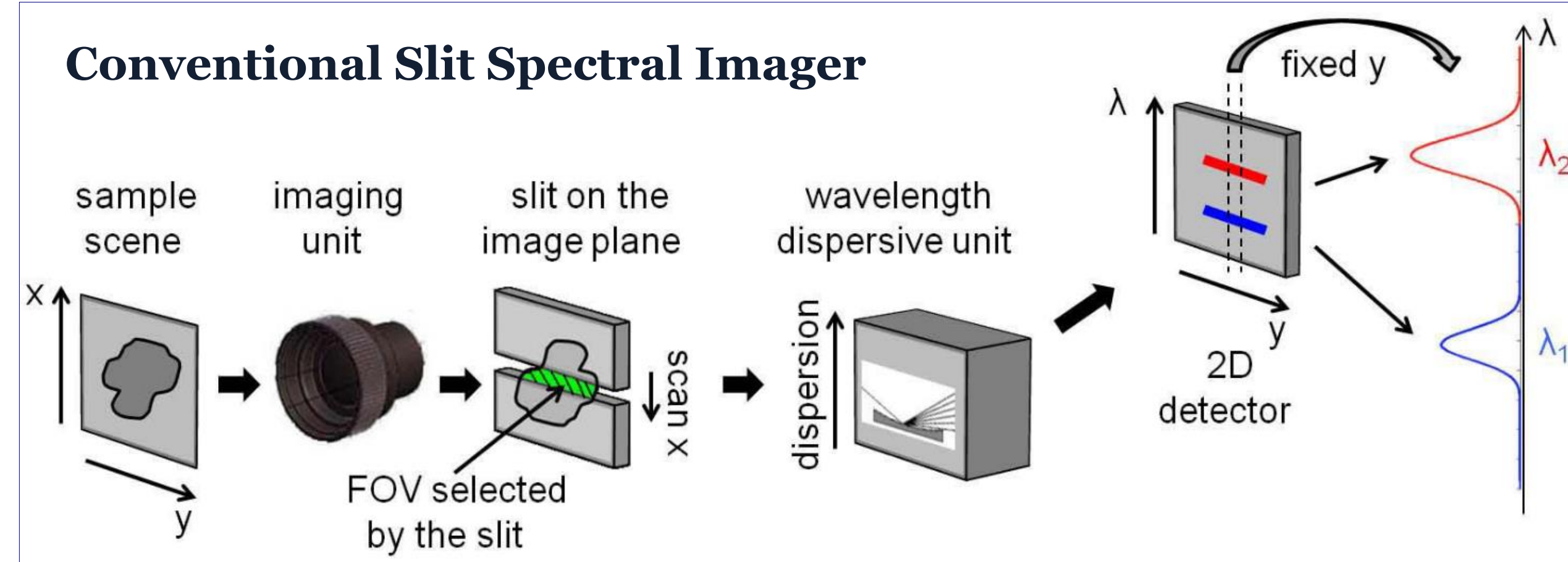
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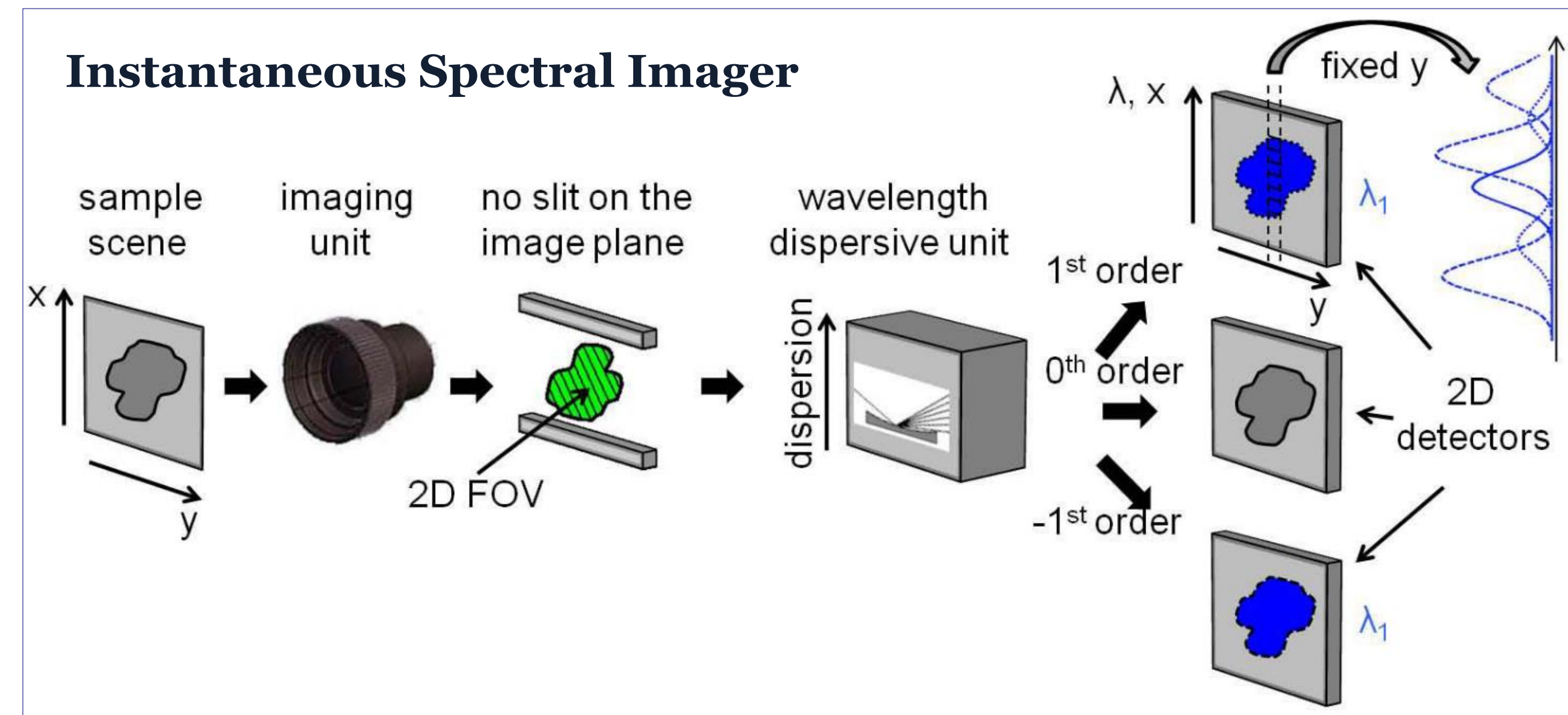


INTRODUCTION

Spectral imaging, simultaneous spectroscopy and photography of a scene, is a fundamental diagnostic technique in physical sciences. Conventional spectral imagers require a scanning process to capture the inherently 3D spatio-spectral scene, which renders them inefficient.



Conventional slit spectrometers require a scanning process by sweeping the slit across the scene, which restricts the field of view to a narrow line, enabling the perpendicular dimension to be used for capturing the spectrum on the detector via dispersion.

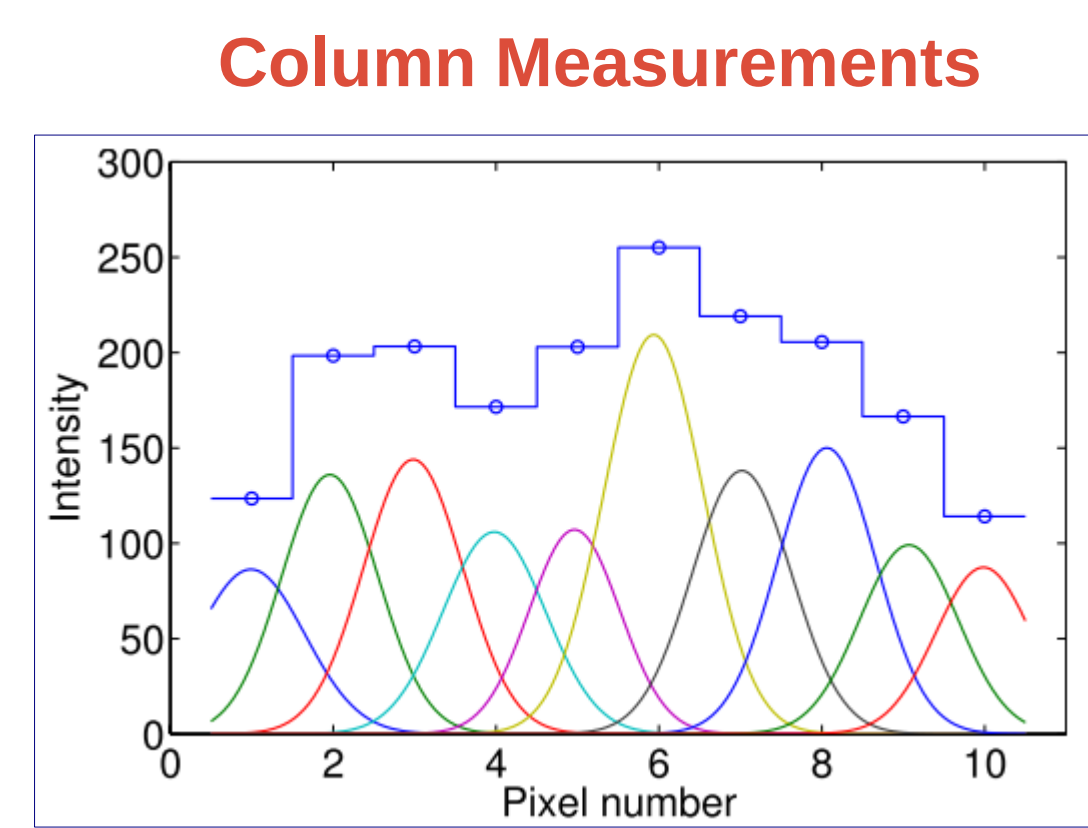
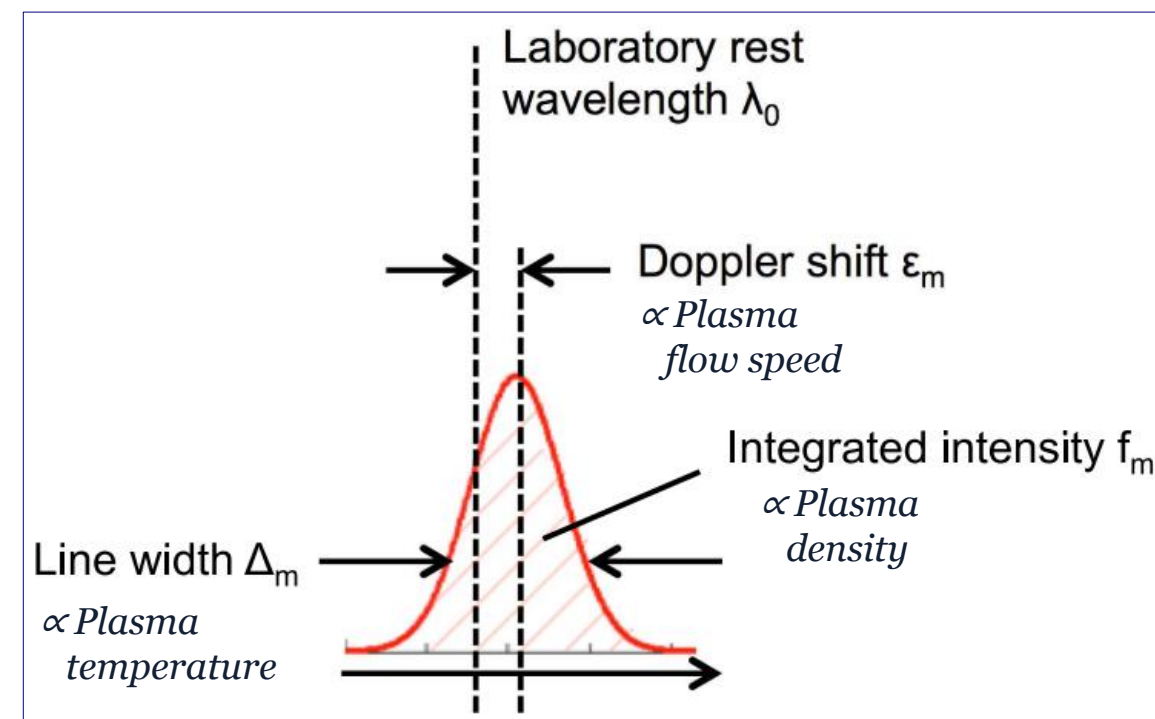


The instantaneous spectral imager removes the slit and does not require a scanning process. This comes with the cost of superposed spectra of the scene along the dispersion direction, to be deconvolved with processing. Multiple diffraction orders can be measured to decrease the ill-posedness of the inverse problem.

PARAMETRIC FORWARD MODEL

Gaussian Spectrum Assumption

- Spectrum consists of Gaussian spectral lines due to thermal broadening, with known central wavelength λ_0 .



Contribution of the spectral line at position m' to pixel m :

$$c_{m'} = \int_{m-1/2}^{m+1/2} \frac{f_{m'}}{\sqrt{2\pi}|a|\Delta_{m'}} e^{-\frac{(x'-m'-a\epsilon_{m'})^2}{2(a\Delta_{m'})^2}} dx'$$

Nonlinear dependence on $\Delta_{m'}$ and $\epsilon_{m'}$.
 a : diffraction order

$$\begin{bmatrix} y_1^{(a)} \\ y_2^{(a)} \\ y_3^{(a)} \\ \vdots \\ y_M^{(a)} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2\pi}|a|\Delta_1} e^{-\frac{(a\epsilon_1)^2}{2(a\Delta_1)^2}} & \frac{1}{\sqrt{2\pi}|a|\Delta_2} e^{-\frac{(1-a\epsilon_2)^2}{2(a\Delta_2)^2}} & \dots & \frac{1}{\sqrt{2\pi}|a|\Delta_M} e^{-\frac{(1-M-a\epsilon_M)^2}{2(a\Delta_M)^2}} \\ \frac{1}{\sqrt{2\pi}|a|\Delta_1} e^{-\frac{(1-a\epsilon_1)^2}{2(a\Delta_1)^2}} & \frac{1}{\sqrt{2\pi}|a|\Delta_2} e^{-\frac{(a\epsilon_2)^2}{2(a\Delta_2)^2}} & \dots & \frac{1}{\sqrt{2\pi}|a|\Delta_M} e^{-\frac{(2-M-a\epsilon_M)^2}{2(a\Delta_M)^2}} \\ \frac{1}{\sqrt{2\pi}|a|\Delta_1} e^{-\frac{(2-a\epsilon_1)^2}{2(a\Delta_1)^2}} & \frac{1}{\sqrt{2\pi}|a|\Delta_2} e^{-\frac{(1-a\epsilon_2)^2}{2(a\Delta_2)^2}} & \dots & \frac{1}{\sqrt{2\pi}|a|\Delta_M} e^{-\frac{(3-M-a\epsilon_M)^2}{2(a\Delta_M)^2}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\sqrt{2\pi}|a|\Delta_1} e^{-\frac{(M-1-a\epsilon_1)^2}{2(a\Delta_1)^2}} & \frac{1}{\sqrt{2\pi}|a|\Delta_2} e^{-\frac{(M-2-a\epsilon_2)^2}{2(a\Delta_2)^2}} & \dots & \frac{1}{\sqrt{2\pi}|a|\Delta_M} e^{-\frac{(a\epsilon_M)^2}{2(a\Delta_M)^2}} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ \vdots \\ f_M \end{bmatrix}$$

Forward model given in matrix-vector form for a single column of measurements.

INVERSE PROBLEM

- The goal in the inversion is to estimate the parameters of the Gaussian spectral lines at all spatial positions within the 2D field of view.

$$\hat{\mathbf{f}}, \hat{\Delta}, \hat{\epsilon} = \arg \min_{\mathbf{f}, \Delta, \epsilon} \{ \underbrace{\|\mathbf{y} - \mathbf{H}(\Delta, \epsilon)\mathbf{f}\|_2^2}_{\text{Data Fidelity}} + \underbrace{\mathcal{R}(\mathbf{f}, \Delta, \epsilon)}_{\text{Regularization / Prior}} \}$$

- An ill-posed multi-frame semi-blind deblurring problem, which is
 - Nonlinear* (in Δ, ϵ) and *nonconvex*; hence, hard to optimize.
- How to choose the regularization / prior term?

DEEP POSTERIOR SAMPLING (DPS)

Score-Based Prior

Diffusion models define the generative process as the *reverse* of the noising process. Define the stochastic differential equation (SDE) for the data noising process $\mathbf{x}(t)$, $t \in [0, T]$.

Forward SDE (data $\rightarrow$ noise)

$$\mathbf{x}(0) \xrightarrow{dx = -\frac{\beta(t)}{2} \mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}} \mathbf{x}(T)$$

Reverse SDE (noise $\rightarrow$ data)

$$\mathbf{x}(0) \xleftarrow{dx = \left(-\frac{\beta(t)}{2} \mathbf{x} - \beta(t) \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \right) dt + \sqrt{\beta(t)} d\mathbf{w}} \mathbf{x}(T)$$

Train a neural network s_θ to approximate the score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$:

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}(\epsilon, 1), \mathbf{x}(t) \sim p(\mathbf{x}(t) | \mathbf{x}(0)), \mathbf{x}(0) \sim p_{\text{data}}} \|\mathbf{s}_\theta(\mathbf{x}(t), t) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}(t) | \mathbf{x}(0))\|_2^2$$

Posterior Sampling

Once the prior is trained, modify reverse SDE to incorporate $\mathbf{y}$:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \xrightarrow[\text{on measurements}]{\text{Condition}} \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)$$

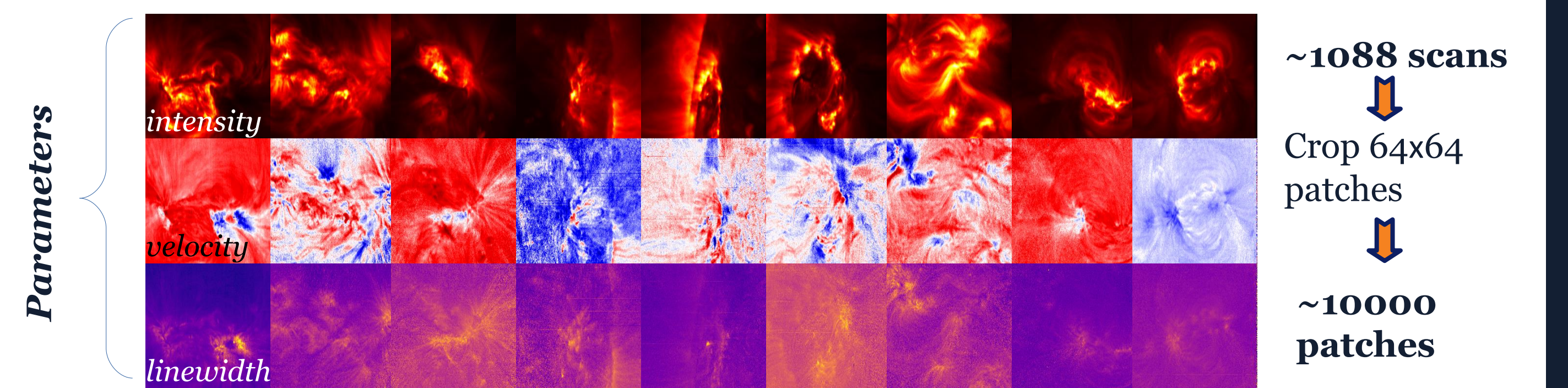
$$p(\mathbf{y} | \mathbf{x}_t) \approx p(\mathbf{y} | \hat{\mathbf{x}}_0), \quad \text{where } \hat{\mathbf{x}}_0 \simeq (\mathbf{x}_t + (1 - \bar{\alpha}(t)) \mathbf{s}_{\theta^*}(\mathbf{x}_t, t)) / \sqrt{\bar{\alpha}(t)}$$

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y}) \simeq \mathbf{s}_{\theta^*}(\mathbf{x}_t, t) - \rho \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\hat{\mathbf{x}}_0)\|_2^2$$

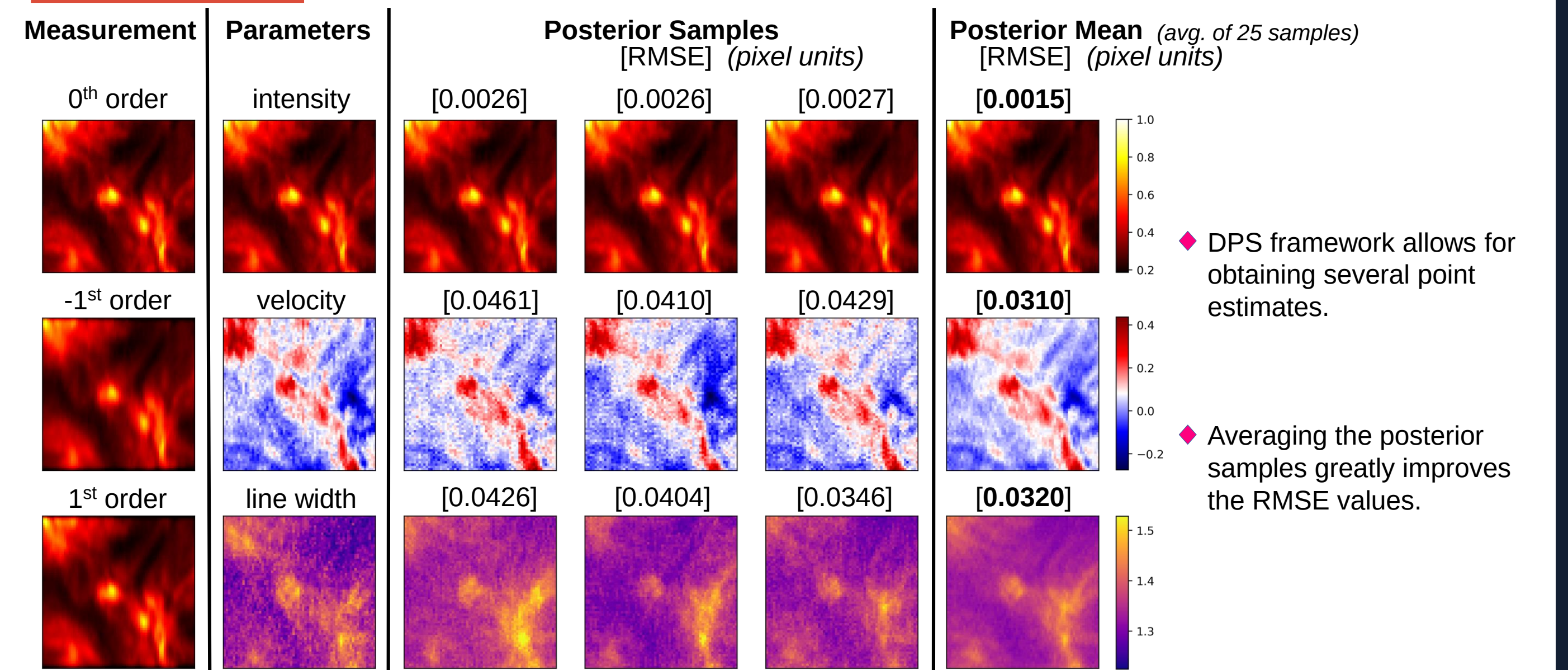
Additional gradient descent step during sampling ensures data consistency.

Dataset

1088 scans of Fe XII 195 Å emission from the Extreme Ultraviolet Imager (EIS) aboard Hinode observatory were selected.



RESULTS



Out-of-Distribution Reconstruction Comparison

